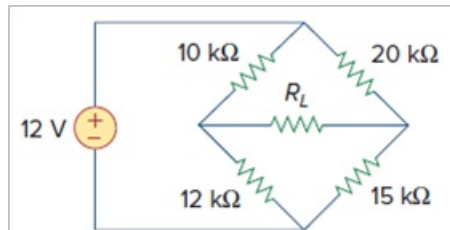


Problem

Consider the circuit in Fig. 4.145. (a) Replace the resistor R_L by a zero resistance ammeter and determine the ammeter reading. (b) To verify the reciprocity theorem, interchange the ammeter and the 12-V source and determine the ammeter reading again.



Step-by-step solution

Step 1 of 4

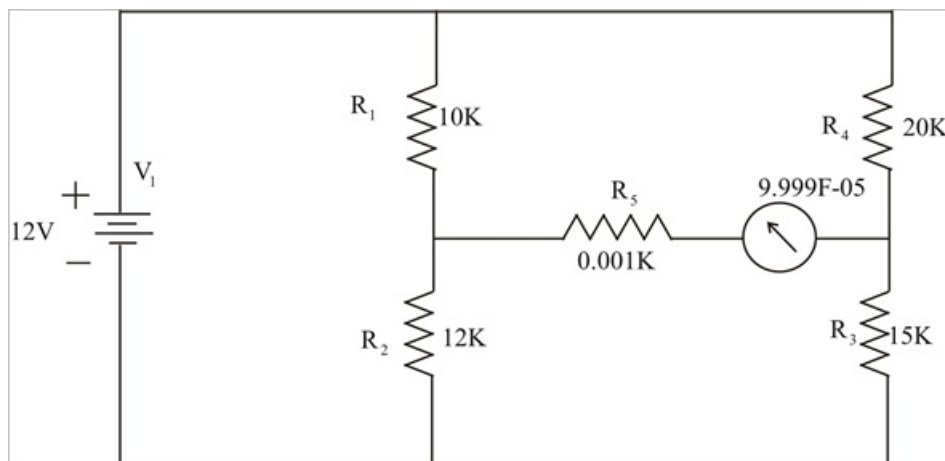
It is easy to solve this problem using P spice.

Step 2 of 4

(a)

The schematic is shown below. We insert iprobe to 1 measure the desired ammeter reading. We insert a very small resistance. In series iprobe to avoid problem. After the circuit is saved and simulated, the current is displaced on iprobe as 99.99 mA.

Step 3 of 4



Step 4 of 4

(b)

By interchanging, the ammeter and the 12 – V voltage source, the schematic is shown below. We obtain exactly the same result as in part(a).

